UV–Vis absorption of conjugated aldehydes: basis set and correlation effects in CIS, ZINDO and TD-B3LYP

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1 Introduction

UV–Vis spectroscopy probes the chromophores of a molecule, that is, its π systems. An α,β-unsaturated aldehyde has two low-lying transitions: an n$\rightarrow\pi^*$ excitation from the oxygen lone pair, which is the lowest in energy but almost forbidden (oscillator strength $f \approx 0$), and a $\pi \rightarrow \pi^*$ excitation, which is allowed and gives the intense band whose maximum, $\lambda_{\max}$, is measured. Extending the conjugation lengthens the delocalised π system and, as in a particle in a box, narrows the HOMO–LUMO gap: the $\pi \rightarrow \pi^*$ transition requires less energy and the band shifts to longer wavelengths (bathochromic shift). This is observed experimentally for propenal, (*E*)-but-2-enal and (*E*)-penta-2,4-dienal (209, 221 and 251 nm [1]).

The simplest *ab initio* method for excited states is configuration interaction singles (CIS) [2], which describes each excited state as the Hartree–Fock ground state with one electron promoted and, like Hartree–Fock, neglects electron correlation. Since correlation stabilises the excited state more than the ground state, CIS is known to overestimate excitation energies, typically by about 1 eV [3]. The hypothesis of this work is that this error comes from the method and not from the basis set: as the basis set grows, the CIS wavelengths should converge, but to values that remain too short. Conversely, time-dependent density functional theory (TD-DFT) includes correlation through the functional and should come much closer to experiment, although TD-B3LYP is known to slightly underestimate the energy of this kind of transition [4], that is, to overshoot $\lambda_{\max}$.

To test it, the $\pi \rightarrow \pi^*$ excitation of the three aldehydes was computed with CIS in five basis sets of increasing size, with the semiempirical ZINDO method [5] (Exercise 6) and, as an extension beyond the exercise, with TD-B3LYP [6, 7].

2 Methods

All calculations were performed with Gaussian 09 [8]. The geometries of propenal, (*E*)-but-2-enal and (*E*)-penta-2,4-dienal were built in their planar s-*trans* conformation and optimized at the HF/3-21G level. On each optimized geometry, the five lowest singlet excited states were computed with CIS in five basis sets of increasing size (3-21G, 6-31G*, 6-31+G*, 6-311+G* and 6-311++G**), with ZINDO and with TD-B3LYP/6-311++G**. All the calculations for each molecule were chained in a single input file and read the optimized geometry from the checkpoint file.

The $\pi \rightarrow \pi^*$ transition was identified, for each method, as the excited state with the largest oscillator strength f, and its wavelength was taken as the calculated absorption maximum, $\lambda = hc/\Delta E$ ($\lambda/\mathrm{nm} = 1239.84/(\Delta E/\mathrm{eV})$). The TD-B3LYP spectra were simulated by broadening each transition with a Gaussian function of width $\sigma = 0.25$ eV in energy, weighted by its f.

3 Results and discussion

Table 1 compares the calculated absorption maxima with experiment. All three methods reproduce the bathochromic shift from propenal to pentadienal, and the small shift of butenal, whose methyl group does not extend the conjugation. However, the size of the error depends strongly on the method.

Table 1: Calculated and experimental $\lambda_{\max}$ (nm) of the $\pi \rightarrow \pi^*$ band, and difference calculated − experimental (Δ). CIS and TD-B3LYP with the 6-311++G** basis set.

		CIS		ZINDO		TD-B3LYP	
Molecule	Exp.	λ	Δ	λ	Δ	λ	Δ
Propenal	209	171.3	−37.7	205.9	−3.1	197.9	−11.1
Butenal	221	173.5	−47.5	218.9	−2.1	205.6	−15.4
Pentadienal	251	209.5	−41.5	263.4	12.4	258.4	7.4

CIS: the error does not come from the basis set. As the basis set grows (Figure 1a), the CIS wavelengths increase by 12–13 nm and then level off: from 6-31+G* onwards they change by less than 1.5 nm. The largest step is the addition of diffuse functions (6-31G* → 6-31+G*), as expected for an excited state, whose electron density is more extended than that of the ground state. Even with the largest basis set, CIS underestimates $\lambda_{\max}$ by 38–48 nm, which corresponds to excitation energies 1.0–1.5 eV too high (Figure 1b), in agreement with the known error of a method without electron correlation [3]. CIS therefore converges with the basis set, but to the wrong answer, as hypothesised.

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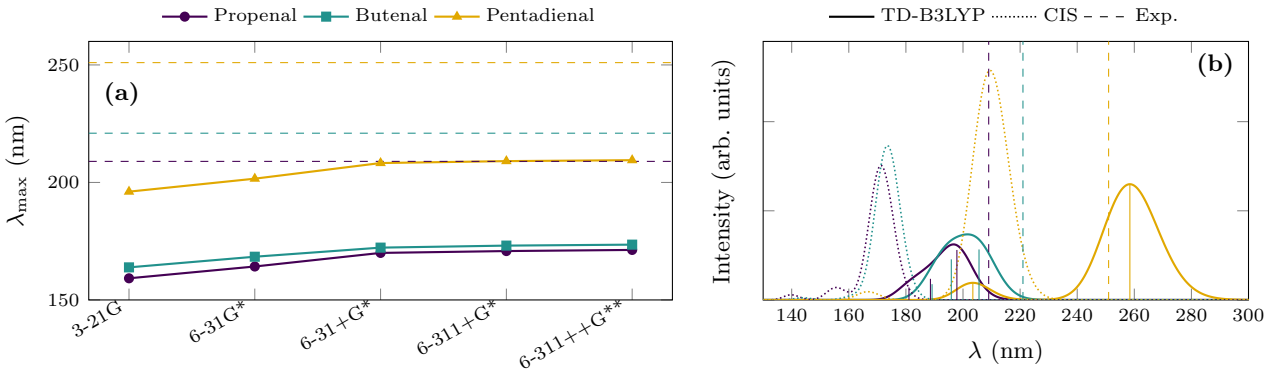


Figure 1: (a) CIS wavelength of the $\pi \to \pi^*$ band against the basis set. (b) Spectra simulated with Gaussian functions ($\sigma = 0.25$ eV; illustrative band shapes, quantitative values in Table 1) from TD-B3LYP (solid; vertical bars, individual transitions of height f) and CIS (dotted), both with 6-311++G**. Dashed lines: experimental maxima. Same colours in both panels.

TD-B3LYP: correlation removes most of the error. TD-B3LYP reduces the error to 7–15 nm (0.1–0.4 eV). Contrary to the hypothesis, however, it does not systematically overshoot λ_{max}: it falls short for propenal and butenal and only overshoots for pentadienal. In other words, its error changes with the length of the π system: TD-B3LYP predicts a red shift of 61 nm from propenal to pentadienal, against 42 nm experimentally. This overestimation of the effect of conjugation is consistent with the tendency of approximate functionals such as B3LYP to over-delocalise the electron density in extended π systems [3].

In TD-B3LYP, the bright state is the second excited state, as expected [1], but its intensity is partly shared with higher states ($f = 0.28$ for propenal and butenal, against 0.65 for pentadienal): in butenal, the third state ($f = 0.23$) shifts the maximum of the simulated band from 206 to 202 nm (Figure 1b). The simulated spectra also reproduce the stronger absorption of the longer chromophore (hyperchromic effect).

ZINDO: right for empirical reasons. ZINDO gives the smallest errors for propenal and butenal (2–3 nm), but this is not a sign of a better physical description: it is a semiempirical method whose parameters were fitted to reproduce experimental spectra [5]. Even so, like TD-B3LYP, it overshoots the red shift of pentadienal.

Two sources of error are common to all the methods. First, the HF/3-21G geometry: Hartree–Fock overestimates the alternation between single and double bonds, which reduces the delocalisation and raises the excitation energy. Second, the calculated values are vertical excitations of an isolated molecule, whereas the experimental maxima are band maxima, probably measured in solution.

4 Conclusion

1. CIS underestimates λ_{max} of the three aldehydes by 38–48 nm (excitation energies 1.0–1.5 eV too high). Increasing the basis set does not fix it: the wavelengths converge from 6-31+G* onwards, but to values that remain far from experiment. The error comes from the lack of electron correlation, not from the basis set.
2. Including correlation through TD-B3LYP reduces the error to 7–15 nm (0.1–0.4 eV). Its error, however, depends on the length of the π system: it falls short for propenal and butenal and overshoots for pentadienal, overestimating the effect of conjugation.
3. ZINDO gives the closest values for the two shorter aldehydes, as expected from a method parametrised against experimental spectra. All the methods reproduce the red shift caused by the conjugation.

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